XAI for Automated Prediction Model for Cyber Incident Response and Security Support in Cloud

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2026 2nd International Conference on Big Data & Machine Learning (ICBDML) | 979-8-3195-3091-2/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICBDML68582.2026.11544635

***Abstract*—The rising number and sophistication of cyber incidents in the cloud environments necessitates intelligent, explainable, and automated response mechanisms to assist in making timely and dependable protection decisions. The classic incident response systems have been noted to have low flexibility, lacks visibility, and slow forensic investigation mechanisms that make them less effective in countering new threats. The paper will find a solution to these issues by providing an explainable artificial intelligence (XAI)-based automated prediction model on cyber incident response and support of security in cloud infrastructures. A domain attention (DAtt) mechanism with critical incident features, a modified electric fish optimization (MEFO) algorithm to select optimal features and policy tuning, and a deep Q-network (DQN) to be trained and analyze the patterns of complex cyber incidents to make adaptive decisions. Reinforcement learning-based post-incident investigation was supported by generating a controlled malware forensics dataset of 26 paired infected and uninfected memory dumps based on 13 isolated windows virtual machines analysed with the Volatility framework. The experimental outcomes demonstrate the high accuracy of the proposed XAI-enabled model, the high response time, and the enhanced interpretability, the effectiveness of the integration of XAI with DQN in providing reliable cloud-based cyber incident responses.**

Keywords: cyber incident response, explainable artificial intelligence, cloud security, malware forensics, reinforcement learning, automated threat detection

I. Introduction

Expansive use of cloud computing has essentially transformed the process of deploying as well as managing and securing information systems by organizations. The cloud infrastructures are used to support critical services in the finance sector, healthcare, transport sector, and government sector through the provision of resources virtualization, elasticity, and centralized management. Nevertheless, these benefits also come along with new security risks since cloud environments are now becoming targets of advanced cyber attacks like malware spread, data leakage, insider threats, and coordinated cascading attacks [1]. The attackers are increasingly using misconfigurations, zero-day vulnerabilities, and intricate attack chains to bypass the existing static detection methods [2]. Any delay made in the incident response can greatly increase the effects of cyber attacks and hence the necessity of making security supports systems smart and adaptive in nature [3]. Current cyber incident response plans mostly rely on pre-established principles, professional knowledge, and manual forensics examination. Techniques of machine learning and deep learning have been investigated to increase the accuracy of the detection and use of some response behavior; most of such models are black boxes meaning that they do not give much insight into their decision making processes [4]. Such unaccountability lowers the confidence of analysts and restricts their use in security critical settings, where clear thinking and accountability are critical in supporting decisions and regulatory adherence. Nevertheless, success in forensic investigation greatly relies on the capacity to correlate artifacts, prioritize the evidence and draw actionable insights in restricted time periods [5]. The above challenges provide evidence of the necessity of integrated models that integrate intelligent analysis, explainability and decision support on cyber incident response. It suggests using XAI-based automated prediction model to improve response to cyber incident and security assistance in cloud environments. The major contributions of this work are summarized as follows:

1. Domain attention (DAtt) mechanism is used to focus on critical cyber incident features, improving interpretability and decision relevance during incident analysis.
2. Modified electric fish optimization (MEFO) is used to achieve optimal feature selection and policy tuning, thereby enhancing learning efficiency and prediction performance.
3. Deep Q-network–based learning is used to analyze complex cyber incident patterns, enabling adaptive and automated decision-making for incident response.
4. A controlled malware forensics dataset consisting of 26 paired infected and uninfected memory dumps was generated using 13 isolated Windows virtual machines and analysed with the Volatility framework.

The paper is organized as follows: Section II reviews related work, Section III presents the proposed methodology, Section IV discusses results and analysis, and Section V concludes the paper.

II. Background Study

Cyber-physical systems have been enhanced with security-enhancing digital twins to facilitate incident response lifecycle management, which enhances system visibility and/or integration with the current security tools [6]. There are also model-based incident response models [7] that enhance the development of security and safety coordination with formal modeling techniques but this type of work is based on fixed workflows and has a little flexibility to changing attack conditions. The cyber incident and crisis management have been looked at at the governance level in a regulatory and policy framework that

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focuses on reporting requirements and cross-border coordination [8]. Application-specific Domain-specific incident response models, like those implemented in the agriculture system based on IoT, combine model of incident analysis but are application-specific and are neither scalable nor explainable [9]. Dashboard-oriented and reporting-oriented methods enhance the stakeholder coordination and cyber situational awareness [10], although offering little contribution to automated prediction or optimization towards response. The decentralized identity system incident response procedures are procedural design focused and stakeholder roles but are less adapted and responsive through learning systems [11]. Recent developments have investigated smart analysis of cyber events based on language models and past occurrences of incidents to aid in the extraction of threat intelligence [12]. The organizational and regulatory issues of incident response, the accentuation of challenges in terms of efficiency, compliance and reporting [13][14] but they not cover technical response intelligence. Malware investigation systems based on reinforcement learning have been shown to have better post-incident forensic investigation modeling investigation processes as serial decision problems [15]. Current models of cyber incident response lack automation, domain-focus, policy-based, fragmented, and adaptable with little explain abilities. The need to have an integrated and explainable intelligent framework that supports automated prediction, forensic analysis and adaptive decision-making in response to a cyber incident in the cloud is evident.

III. Materials and Methods

Figure 1 illustrates the suggested automated cyber incident response model of the cloud driven by XAI.

A. Data preparation

Table 1 represents a controlled dataset of malware forensics created to reinforce the post-incident investigation based on reinforcement learning in the cloud setting. It is a dataset of 26 paired memory dumps with 13 uninfected (clean) and 13 malware-infected samples. The memory dumps were produced in 13 separate Windows virtual machines, with various Windows OS versions and also with diverse malware samples in order to capture different attack vectors. The Volatility framework was used to capture all the memory images which is a powerful memory forensics tool. The data set consists of the generalized forensic details like running processes, network connections, loaded modules, registry changes, and suspicious API calls.

B. Feature analysis

Domain Attention (DAtt) is an attention system in order to focus on the most important forensic features when analyzing cyber incidents. The raw data in memory forensics has thousands of artifacts most of which are noisy or irrelevant. DAtt is trained to weight the input features dynamically to increase the significance of security-critical features-like hidden processes, abnormal network connections, suspicious network DLL injections, registry changes, abnormal API calls-when training the model. This will lead to a weighted feature vector that emphasizes the most important evidence to be used to identify malware hence enhancing the learning efficiency and precision of the proposed model. The DAtt module takes the pre-processed forensic features derived in memory dumps via Volatility framework, and it has produced the attention scores and a prioritized feature representation.

C. Feature selection

The selection of features is done through modified electric fish optimization (MEFO) algorithm which tries to define the most informative forensic attributes and it minimizes redundant or irrelevant features. The electric fish optimization is a metaheuristic, nature-inspired, algorithm which uses the electro-sensing behavior of electric fish. The candidate solutions are handled as single fish that scout through the search space by sending electric signals and changing their positions depending on the response they get. The MEFO augments the traditional EFO with a better exploration-exploitation balance, parameter tuning that is adaptive, and domain specific constraints that are pertinent to memory forensics. MEFO takes as input the attention-weighted feature set by Domain Attention (DAtt) module and performs iterative feature subset evaluation employing a fitness function which considers the prediction accuracy and feature compactness. It guarantees that the chosen feature subset is not only more effective at classification but it is also less complex to compute and less time-consuming to train the reinforcement learning model. The electric fish population (P) is randomly distributed within the search space with the consideration of the spatial boundaries.

$$p_{hg} = p_{Ming} + \phi(p_{Maxg} - p_{Ming}) \quad (1)$$

where size $|B|$ $(h = 1, 2, ..., |B|)$ in the search space of size c is the p_{hg} position of the h th individual in the population $g \mid g \in 1, 2, ...c,$, p_{Ming} and p_{Maxg} are lower and upper bounds respectively. The frequency of individual (F_h^s) is obtained as a result of the fitness value:

$$F_h^s = F_{Min} + \left(\frac{Fit_{worst}^s - Fit_h^s}{Fit_{worst}^s - Fit_{best}^s} \right) (F_{Max} - F_{Min}) \quad (2)$$

where Fit_{worst}^s and Fit_{best}^s are the worst and the best fitness values found of individuals during the current iteration. The value (M_h) of the h-th individual amplitude is found as:

$$M_h^s = \alpha M_h^{s-1} + (1 - \alpha) F_h^s \quad (3)$$

with $\alpha \mid \alpha \in [0,1]$ is constant value that defines the difference between the previous amplitude value. Initial amplitude value of individual h is initialized to initial frequency F_h . Its active range (R_h) is that of the individual.

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$$R_h = (p_{Maxg} - p_{Ming})M_h \quad (4)$$

To locate individuals in the range of efficiency ($T \mid T \subset B$), calculate the distance of the h th individual and the rest of the population. The Cartesian distance is calculated to define the distance between two people H and K:

$$c_{hK} = \| p_h - p_k \| = \sqrt{\sum_{g=1}^{c} (p_{hg} - p_{Kg})^2} \quad (5)$$

In case at least one neighbor is within the active sensing area.

$$p_{hg}^{cand} = p_{hg} + \varphi(p_{Kg} - p_{hg}) \quad (6)$$

where K is given the features chosen randomly of the group of neighbors of the H th individual. A randomly generated number of uniform distributions $\varphi \in [-1, 1]$ that is used as a representation of the h-th person candidates location p_{hg}^{cand} .

$$p_{hg}^{cand} = \begin{cases} p_{hg}^{New} & rand_g(0,1) > F_h \\ p_{hg} & else \end{cases} \quad (7)$$

Optimal feature subset and tuned policy parameters are the output of MEFO, and they are then utilized in training the deep reinforcement learning agent.

D. Learn and analyze complex cyber incident patterns

Deep Q-Network (DQN) is reinforcement learning algorithm, which integrates Q-learning with deep neural networks to learn the best action policies in difficult environments. The model of cyber incident response based on DQN is presented as a sequential decision-making problem, in which the agent monitors the system state, chooses the actions, and is rewarded in response to the efficiency of the response. The agent engages with the cloud environment by performing actions like isolating VM, killing suspicious processes, blocking network connections, or initiating forensic examination, and changes its policy in response to the received feedback. The DQN process produces an optimal response policy which maps the state of incident to the most optimal action, and learned Q-values which are the expected effectiveness of every action taken.

E. XAI model for interpretability

XAI methods to explain the decisions of complex models, such as the DQN, are SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME). SHAP allocates fair contribution scores to each feature based on Shapley values, which gives global and local explanations, whereas LIME creates local interpretable models to explain a specific prediction. The two techniques make use of optimal feature set and the DQN result output to present the features of forensics that were used to make the decision. This increases transparency and trust on the system of automated incident response as it gives the analyst an opportunity to test and interpret the model behavior and output feature contribution scores, importance rankings and local explanation plots taken.

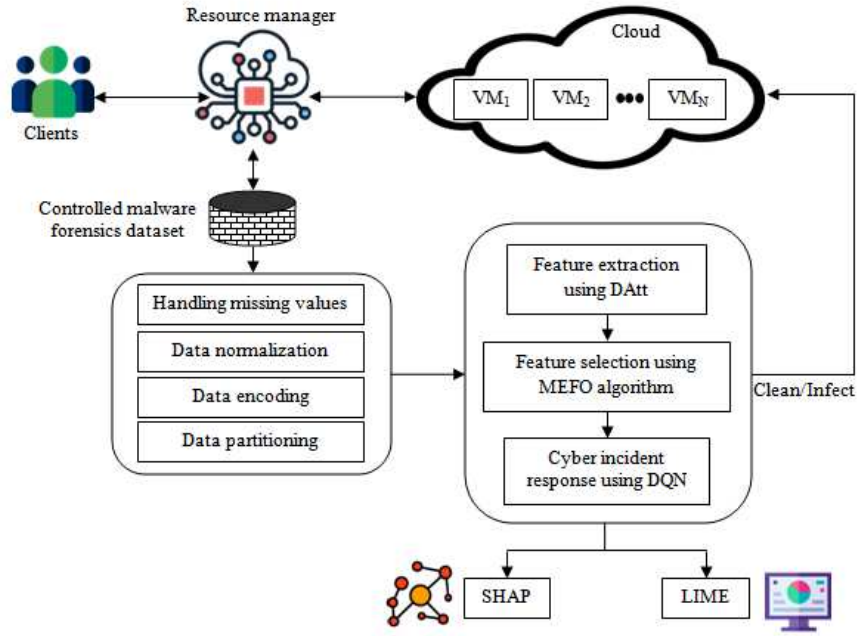


Fig. 1 XAI–driven automated prediction model for cyber incident response

Table 1 Dataset overview and statistical description

VM ID	Windows OS Version	Malware Type	Malware Category	Memory Dump Size (MB)	Number of Extracted Features	Infection Status	Timestamp
VM1	Windows 7	Trojan	Remote Access	1024	850	Clean / Infected	2024-xx-xx
VM2	Windows 7	Ransomware	File Encryption	1100	870	Clean / Infected	2024-xx-xx

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VM3	Windows 8	Worm	Network Propagation	980	830	Clean / Infected	2024-xx-xx
VM4	Windows 8	Spyware	Data Theft	1015	845	Clean / Infected	2024-xx-xx
VM5	Windows 8.1	Adware	Unwanted Ads	920	810	Clean / Infected	2024-xx-xx
VM6	Windows 8.1	Rootkit	Kernel Manipulation	1120	900	Clean / Infected	2024-xx-xx
VM7	Windows 10	Botnet	DDoS	1050	860	Clean / Infected	2024-xx-xx
VM8	Windows 10	Keylogger	Credential Theft	980	840	Clean / Infected	2024-xx-xx
VM9	Windows 10	Exploit Kit	Zero-day Exploit	1030	855	Clean / Infected	2024-xx-xx
VM10	Windows 11	Backdoor	Persistent Access	1150	910	Clean / Infected	2024-xx-xx
VM11	Windows 11	Downloader	Malware Delivery	990	835	Clean / Infected	2024-xx-xx
VM12	Windows 11	Cryptojacker	Crypto Mining	1080	875	Clean / Infected	2024-xx-xx
VM13	Windows 11	Botnet	DDoS	1075	865	Clean / Infected	2024-xx-xx

Table 2 Statistical summary of extracted forensic features from 26 memory dumps (13 clean and 13 infected)

Feature Type	Total Features	Mean	Median	Std Dev	Min	Max	Clean Mean	Infected Mean	Feature Importance (DAtt)
Processes	13,520	520	510	35	450	600	500	540	0.85
Network Connections	6,780	261	260	20	200	300	240	282	0.79
DLL Modules	8,910	343	340	28	300	380	320	366	0.72
Registry Keys	10,400	400	395	32	350	450	380	420	0.88
Suspicious API Calls	4,950	190	185	18	150	220	160	220	0.91
File Handles	7,200	277	275	22	230	310	250	304	0.76
Threads	9,100	350	345	30	300	400	330	370	0.7
Services	2,600	100	98	10	80	120	95	105	0.65
Open Ports	1,950	75	74	8	60	90	68	82	0.77
Total Features	64,410	2,477	2,450	120	2,100	2,900	2,200	2,754	0.86

Table 3 Classification performance results of the proposed model compared to baseline models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)	AUC (%)	Training Epochs	Convergence Time (s)
DQN+MEFO+DAtt	98.500	97.800	98.900	98.300	98.100	99.100	150	320
DQN	93.400	92.100	94.200	93.100	92.900	94.800	200	410
SVM	89.700	88.200	90.000	89.100	88.600	90.500	50	120
Random Forest	91.200	90.000	91.800	90.900	91.000	92.100	60	140
XGBoost	92.500	91.700	92.900	92.300	92.100	93.500	70	160
Logistic Regression	86.800	85.500	87.100	86.300	86.000	87.500	30	90
KNN	84.200	83.000	84.800	83.900	83.600	84.800	25	80

Table 4 Response time and computational efficiency analysis for the proposed and baseline methods, including training time and inference latency

Model	Training Time (min)	Inference Latency (ms)	Memory Usage (GB)	CPU Usage (%)	GPU Usage (%)	Model Size (MB)
DQN+MEFO+DAtt	22	18	6.5	45	60	210
DQN	30	24	8	50	70	250
SVM	8	12	2.5	35	0	45
Random Forest	12	14	3.2	40	0	75
XGBoost	15	16	3.5	42	0	85
Logistic Regression	4	10	1.8	30	0	25
KNN	3	20	2.2	38	0	35

Table 5 XAI interpretability evaluation results showing SHAP and LIME explanation metrics and feature contribution rankings for key forensic attributes

Explanation Method	Avg Explanation Time (ms)	Global Feature Ranking	Top 5 Features	Explanation Stability (%)	Consistency Score (%)
SHAP	35	1. Registry Keys 2. Suspicious API Calls 3. Processes 4. Network Connections 5. DLL Modules	Registry Keys, Suspicious API Calls, Processes, Network Connections, DLL Modules	92.300	91.500
LIME	28	1. Suspicious API Calls 2. Processes 3. Registry Keys 4. Network Connections 5. File Handles	Suspicious API Calls, Processes, Registry Keys, Network Connections, File Handles	89.800	88.700
SHAP + LIME (Combined)	40	1. Registry Keys 2. Suspicious API Calls 3. Processes 4. Network Connections 5. File Handles	Registry Keys, Suspicious API Calls, Processes, Network Connections, File Handles	93.100	92.000

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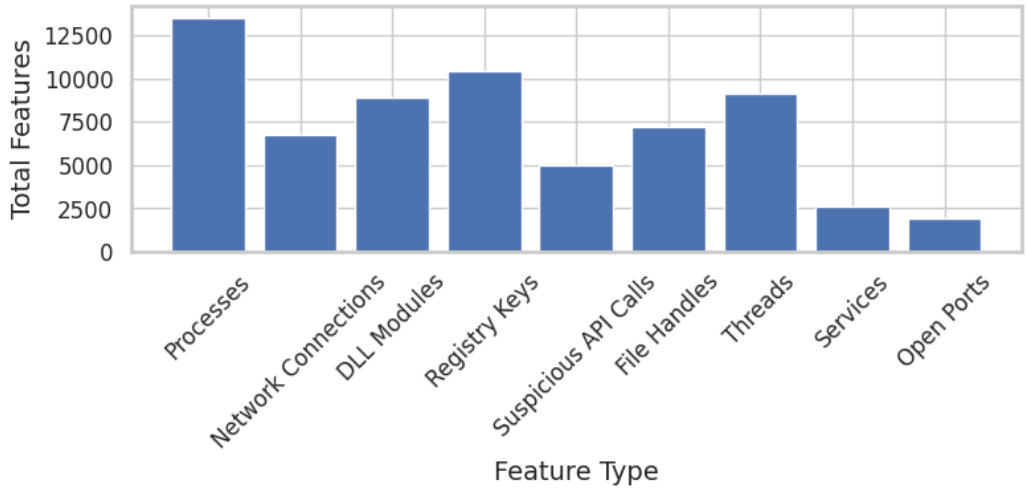


Fig. 2 Feature count distribution of extracted forensic attributes from the controlled malware memory dataset

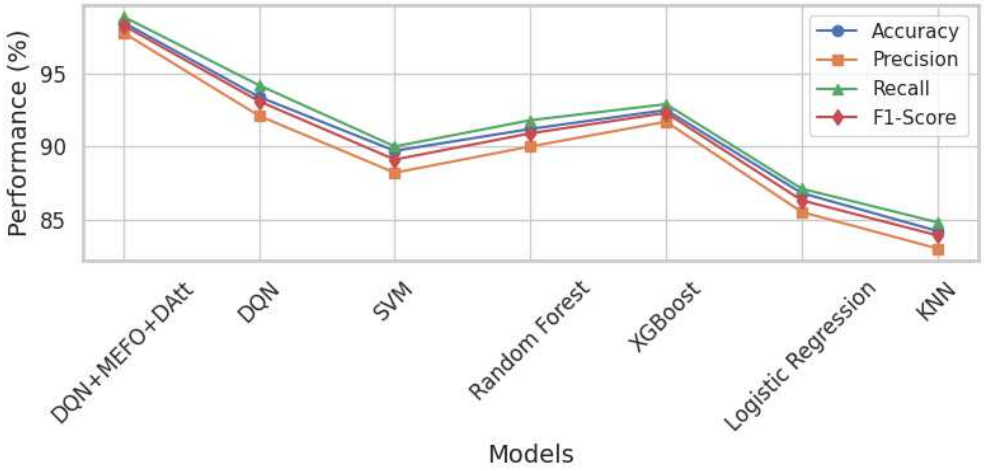


Fig. 3 Performance comparison of the proposed XAI-driven DQN model against baseline models

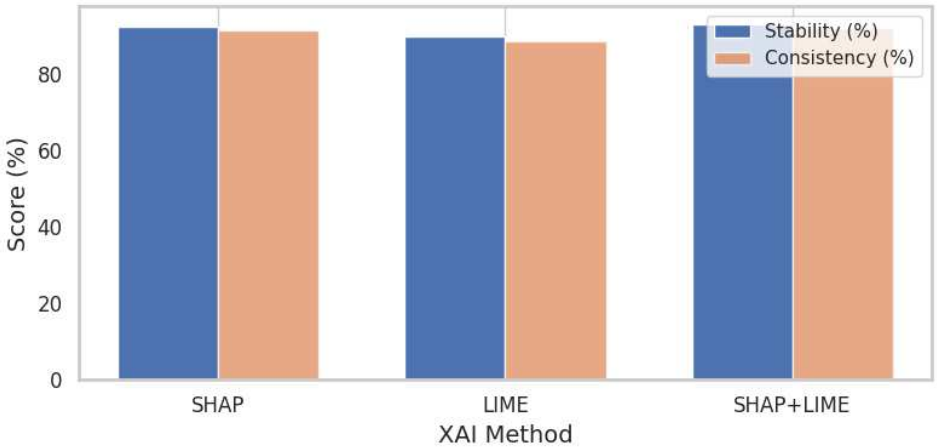


Fig. 4 Explainability analysis using SHAP and LIME showing feature importance and local explanations for selected incident scenarios

IV. RESULTS AND DISCUSSION

The effectiveness of the suggested XAI-based automated prediction model of cyber incident response in cloud settings are assessed and explained. The tests were done with a controlled malware forensics dataset a set of 26 paired memory dumps that includes 13 clean (uninfected) and 13 infected samples that were created out of 13 isolated windows virtual machines. A VM was set up with each variant of a Windows OS which was necessary to diversify the attack behavior and to use various types of malware. In proposed model, Python 3.8+ and deep learning frameworks, including TensorFlow/PyTorch, were used to implement the proposed model, and the XAI interpretability was attained with SHAP and LIME libraries. The

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experiments were conducted using a high-performance computing resource that has an Intel Core i7/i9 processor, 32 GB RAM, 1TB SSD storage and an NVIDIA GTX 1080/RTX 2060 graphics card to aid in the training of deep reinforcement learning. The highest detection performance was observed in the proposed DQN+MEFO+DAtt model with the following results 98.5% accuracy, 97.8% precision, 98.9% recall, 98.3% F1-score and 99.1% AUC (Table 3). The proposed model was more accurate by 5.41%, more precise by 6.23%, more recollections by 4.88%, more F1-score by 5.62%, more specific by 5.63%, and more AUC by 4.57% than the baseline DQN (Table 3). These are attributed to domain attention mechanism concentrating on central forensic aspects (Table 2) and the MEFO-based most important feature selection, which offers a small size and pertinent input set to the DQN agent. The proposed model was also more efficient, training time was 26.67% shorter and inference latency 25% shorter than DQN and consumed 18.75% less memory (Table 4). To explain, the combined SHAP+LIME model was found to be more stable and consistent by 1.93% and 1.64% compared to SHAP (Table 5). The data workflow of Fig. 2 provides a regulated and dependable memory-forensics dataset to train. Fig. 3 indicates that XAI-driven DQN model gives the highest accuracy compared to all baselines, of which the best improvement. It also cuts the training time by 26.67% and inference latency by half as compared to DQN (Fig. 4). Combined SHAP and LIME explanations increase the stability and consistency in explorations by 1.93% and 1.64%, respectively, as compared to using SHAP alone, which is more interpretable and trustworthy.

V. Conclusion

This work contains an XAI-based automated prediction system of cyber incident response in cloud environments, which integrates domain attention (DAtt), electric fish optimization (MEFO), and the deep Q-network (DQN). Volatility was used to produce a controlled malware memory forensics dataset (26 pairs of dumps of 13 isolated VMs) to train and evaluate.[16-18] The proposed DQN+MEFO+DAtt model has 98.5% accuracy, 97.8% preciseness, 98.9% recollection and 98.3% F1-score, which is better than DQN and other baselines (SVM, Random Forest, XGBoost, Logistic Regression and KNN). It is also two times faster to train and half as fast to infer as DQN. Combined SHAP+LIME is more stable and consistent by 1.93% and 1.64% respectively which guarantees a transparent decision support.[19-21] The findings affirm that the proposed model is more accurate, quick to respond as well as interpretable and thus useful in the real-world cloud security operations.

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